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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools (BL4,BL5)

Assessment Tool Weightage: 40% **Total Marks: 30**

QUESTIONS: 3

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

SOLUTIONS TO CONSTRAINT-BASED PROBLEMS

Question 1: Connectivity Solution for a Remote Rural Branch Office

1. Understanding of the Problem & Constraints

- No high-speed wired internet (no fibre/leased-line availability) in the rural location.
- Must support cloud-based applications, video conferencing, and secure communication with headquarters (HQ).
- Limited connectivity budget — CAPEX/OPEX must stay low.
- Business operations must be uninterrupted — reliability and uptime are critical.
- Solution must be scalable as the branch grows.

2. Problem Decomposition

- Sub-problem A: Selecting a primary internet access technology suited to a rural, no-wired-infrastructure area.
- Sub-problem B: Providing a backup/failover link to guarantee uninterrupted operation.
- Sub-problem C: Choosing the internal networking devices (router, switch, Wi-Fi) for the office LAN.
- Sub-problem D: Designing the security mechanisms for safe communication with HQ and cloud services.
- Sub-problem E: Ensuring the design is cost-effective now and scalable later.

3. Proposed Solution

Primary Internet Link: A satellite broadband service (e.g., a LEO satellite service such as Starlink Business) or a fixed-wireless 4G/5G connection. Satellite/LTE needs no trenching or cabling, can be installed quickly in rural terrain, and provides 50–200 Mbps — sufficient for cloud apps and video conferencing.

Backup Link: A secondary 4G/5G LTE router configured for automatic failover, so if the primary satellite/wireless link degrades, traffic seamlessly switches over, keeping operations uninterrupted.

Core Networking Devices: A dual-WAN SD-WAN router/firewall appliance (handles automatic failover and load balancing), a managed Layer-2/3 switch for the LAN, and a Wi-Fi 6 access point for wireless coverage inside the office.

Security Mechanisms: A site-to-site IPsec VPN tunnel to HQ for encrypted communication; a Unified Threat Management (UTM) / next-generation firewall with IDS/IPS for perimeter protection; TLS/HTTPS enforced for all cloud and video-conferencing traffic; multi-factor authentication (MFA) for remote logins; endpoint antivirus and centralized patch management.

Power/Reliability Backup: A UPS with a few hours of backup runtime protects against local power outages common in rural areas, and dual-WAN failover protects against link outages.

4. Justification Against Constraints

Bandwidth: Satellite/4G-5G links deliver enough throughput (tens to hundreds of Mbps) for cloud SaaS tools and HD video conferencing, unlike traditional dial-up/leased lines that are unavailable here.

Reliability: The dual-WAN failover design (satellite/wireless primary + LTE backup) and UPS ensure the office is never fully offline, satisfying the 'uninterrupted operations' requirement.

Cost: Satellite/LTE services avoid the very high cost of laying fibre/leased lines to a remote site; SD-WAN and UTM appliances bundle routing, firewalling and VPN into a single low-maintenance device, reducing both CAPEX and OPEX.

Future Scalability: Satellite and LTE plans can be upgraded to higher bandwidth tiers on demand; SD-WAN architecture allows additional branch sites to be added to the same secure fabric; the switch and AP can be expanded with additional ports/APs as headcount grows, without redesigning the network.

5. Creativity / Innovation Note

Using SD-WAN with LEO satellite as the primary link (instead of a traditional single-ISP leased line) is an innovative, modern approach purpose-built for rural connectivity — it removes dependency on physical last-mile infrastructure entirely while still meeting enterprise-grade bandwidth and security needs.

Question 2: Campus-Wide Wireless Network Design for a University

1. Understanding of the Problem & Constraints

- More than 3,000 students need simultaneous wireless access.
- Coverage spans classrooms, laboratories, libraries, and hostels — each with different user density.
- Signal interference must be minimized across densely packed buildings.
- Access must be secure (only authorized students/staff).
- Only a limited number of access points (APs) can be deployed due to budget.

2. Problem Decomposition

- Sub-problem A: Estimate user density per zone (classrooms/labs = high density, libraries = medium-high, hostels = medium/low) to plan AP capacity.

- Sub-problem B: Decide AP hardware/technology that maximizes users-per-AP so fewer APs are needed.
- Sub-problem C: Plan physical AP placement to maximize coverage and minimize interference.
- Sub-problem D: Select frequency bands and channel plan to avoid co-channel interference.
- Sub-problem E: Design authentication and access-control mechanisms for security.

3. Proposed Solution

AP Technology: Deploy Wi-Fi 6 (802.11ax) access points. Wi-Fi 6 uses OFDMA and MU-MIMO to serve many simultaneous clients per radio far more efficiently than older standards, so each AP can comfortably support 60–80 concurrent users — directly reducing the total number of APs required to serve 3,000+ students.

AP Placement: Perform a predictive RF site survey/heatmap before installation. Place high-capacity, ceiling-mounted APs with directional coverage in high-density zones (lecture halls, libraries, labs); use standard omnidirectional APs spaced along corridors for hostels, where device density per room is lower. Stagger AP locations building-to-building to avoid overlapping cells that would otherwise cause co-channel interference.

Frequency Band Selection: Operate dual-band (2.4 GHz + 5 GHz) with band-steering to push capable devices to the less congested 5 GHz band. Use non-overlapping channels (1, 6, 11) on 2.4 GHz, and wider 20/40 MHz channels on 5 GHz. Where budget allows, use Wi-Fi 6E-capable APs in the highest-density buildings (library, main lecture halls) to tap the largely interference-free 6 GHz band.

Centralized Management: Use a wireless LAN controller (WLC) or cloud-based controller with Radio Resource Management (RRM) that automatically tunes channel and transmit power on each AP to reduce interference as usage patterns change.

Authentication & Security: Implement WPA3-Enterprise with 802.1X authentication tied to a RADIUS server linked to the university's student/staff directory, so each user connects with individual credentials. Separate SSIDs/VLANs for students, faculty, and guests, with a captive-portal, isolated guest VLAN for visitors.

4. Justification — Balancing Coverage, Performance, Security & Cost

Coverage: Site-survey-driven placement with directional APs in dense zones and spaced omnidirectional APs in hostels ensures the limited AP count still covers all required areas.

Performance: Wi-Fi 6's OFDMA/MU-MIMO plus dual/tri-band operation lets each AP serve more users at higher throughput, meeting the 3,000+ simultaneous-user requirement without needing one AP per few users.

Security: WPA3-Enterprise + 802.1X/RADIUS ensures only authenticated members access the academic network, while VLAN segmentation isolates guest traffic from the academic network.

Cost: Because Wi-Fi 6 APs handle more users each, the university needs fewer physical APs than it would with older Wi-Fi standards — directly respecting the constraint on the limited number of APs that can be purchased/deployed.

5. Creativity / Innovation Note

Concentrating Wi-Fi 6E (6 GHz) APs only in the highest-density buildings, while using standard Wi-Fi 6 elsewhere, is a cost-optimized hybrid deployment — it puts the newest, interference-free spectrum where it delivers the most value instead of uniformly over-provisioning the whole campus.

Question 3: Secure Network Architecture for a Financial Institution

1. Understanding of the Problem & Constraints

- Must strengthen cybersecurity while employees still need access to sensitive customer data and online banking services.
- Must comply with strict regulatory/security policies (e.g., PCI-DSS/ISO 27001-type requirements).
- Security measures must not significantly degrade network performance.
- Operational costs must not increase significantly.

2. Problem Decomposition

- Sub-problem A: Segment the network so sensitive systems are isolated from general traffic.
- Sub-problem B: Deploy firewalls at the right layers to filter traffic without bottlenecking performance.
- Sub-problem C: Choose authentication mechanisms strong enough for financial data access.
- Sub-problem D: Add intrusion detection/prevention and monitoring for early threat detection.
- Sub-problem E: Ensure the overall design maps to compliance requirements at a manageable cost.

3. Proposed Solution

Network Segmentation: Apply VLAN-based zero-trust segmentation: (1) a Core Banking Zone hosting transaction/database servers, (2) a DMZ for customer-facing online-banking web/application servers, (3) an Employee/Corporate LAN, and (4) a Guest/IoT zone. Internal firewalls or Layer-3 ACLs control traffic between zones so a compromise in one zone cannot directly reach another.

Firewall Deployment: A Next-Generation Firewall (NGFW) with deep packet inspection at the internet perimeter (protecting the DMZ), plus a second internal firewall between the DMZ/employee LAN and the Core Banking Zone. Hardware-accelerated NGFWs keep throughput high so security does not bottleneck performance.

Authentication Methods: Multi-factor authentication (MFA) for every employee accessing sensitive systems or online-banking back-ends; role-based access control (RBAC) so employees only see data relevant to their role; 802.1X network access control so only authorized devices can join the internal LAN; certificate-based/PKI authentication and a Privileged Access Management (PAM) solution for administrator accounts.

Intrusion Detection/Prevention: A signature- and anomaly-based IDS/IPS at the perimeter and at the Core Banking Zone boundary, feeding into a centralized SIEM (Security Information and Event Management) system for real-time correlation, alerting, and audit-log retention required by regulators.

Data Protection: Encryption in transit (TLS 1.2+) for all customer and banking traffic, encryption at rest (AES-256) for stored customer data, and Data Loss Prevention (DLP) policies to stop sensitive data leaving the network unencrypted.

4. Justification — Security, Performance, Compliance & Budget

Security: Segmentation plus layered firewalls, MFA, RBAC and IDS/IPS create defense-in-depth, so a breach in one zone (e.g., the DMZ) is contained and cannot reach core banking systems or customer databases.

Performance: Hardware-accelerated NGFWs and a segmentation design that filters traffic only at zone boundaries (rather than inspecting every packet everywhere) keep latency low for time-sensitive banking transactions.

Compliance: Segmentation of card/customer-data environments, encryption, MFA, and centralized audit logging via SIEM directly map to common financial-sector compliance frameworks (e.g., PCI-DSS, ISO 27001), satisfying regulatory obligations.

Budget: Reusing VLAN segmentation on existing switches (rather than fully separate physical networks), and using a cloud- or subscription-based SIEM instead of large on-premise log infrastructure, keeps additional CAPEX/OPEX controlled while still meeting security and compliance goals.

5. Creativity / Innovation Note

Combining zero-trust micro-segmentation with a PAM layer specifically for administrator accounts is a targeted innovation: